

EMİR ÇABALAK

AI / Machine Learning Intern | Deep Learning & Computer Vision

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emircabalak

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EXPERIENCE

AI Intern

Jan 2026 – Jan 2026

Gap İnşaat

Istanbul

- Built a deep learning model that reads 3D Revit building views to identify structural elements and auto-generate material lists in Excel, replacing manual takeoff work.
- Added a cost-estimation layer that combines the extracted materials with current market prices to give a quick project budget estimate.

AI Intern

Feb 2025 – Aug 2025

Acunmedya Academy

Remote

- Completed a hands-on AI program, building machine learning projects across classification, regression, clustering, and ensemble methods – code at github.com/emircabalak/Acunmedia_Academy_Assignments.
- Trained and tuned Random Forest, SVM, and gradient boosting models, reaching 90%+ accuracy on multiple datasets using GridSearchCV and ROC-AUC evaluation.

AI Intern

Jan 2025 – Feb 2025

Mavinci Bilişim

Ankara

- Built CNN-based digit-recognition models and trained them on MNIST and a custom dataset, comparing architectures to improve accuracy.
- Handled annotation and preprocessing for image datasets of 10,000+ samples, preparing clean training data for computer vision tasks.

AI Intern

Jun 2024 – Jul 2024

OSTİM Technical University

Ankara

- Built regression models in Python and Scikit-learn, improving accuracy through data cleaning and feature engineering.
- Learned the core ML workflow end to end: data preparation, model training, and evaluation.

PROJECTS

DocuMind – Production RAG Platform

2025

LangGraph · FastAPI · ChromaDB · github.com/emircabalak/DocuMind

- Built a full-stack document Q&A platform on a 4-node LangGraph agent pipeline (query routing, hybrid retrieval, validation, synthesis), combining BM25 and semantic search via Reciprocal Rank Fusion with CrossEncoder re-ranking.
- Added async document ingestion with Celery + Redis, conversation memory, RAGAS evaluation, and a live analytics dashboard; deployed via Docker Compose.

Unsupervised Network Intrusion Detection System

2025

Deep Autoencoder · KDD Cup 99

- Trained a Deep Autoencoder exclusively on benign network traffic to detect zero-day attacks at 99% recall, eliminating reliance on labeled attack data.
- Implemented reconstruction-error thresholding for high-precision unsupervised anomaly detection.

Confidence-Aware BEVNet

2025

3D Object Detection · KITTI

- Integrated a confidence-score head into a standard Bird's Eye View architecture, enabling quantifiable reliability assessment for 3D detections.
- Improved decision-making safety in autonomous-driving pipelines by filtering low-confidence predictions through uncertainty estimation.

Industrial Acoustic Anomaly Detection

2025

Predictive Maintenance · Unsupervised Learning

- Benchmarked LOF, One-Class SVM, and Deep Autoencoder models on industrial machine-audio recordings to identify the optimal fault-detection approach.
- Built an unsupervised anomaly-detection pipeline trained solely on nominal operating data, removing dependency on labeled fault samples.

ABOUT

Third-year AI Engineering student at OSTİM Technical University, seeking an AI / Machine Learning internship. Experience across the full ML workflow – data, modeling, evaluation, and deployment – through four internships and self-built projects in deep learning, computer vision, and LLM-based systems (RAG, anomaly detection, 3D perception).

SKILLS

Programming Languages

Python
R
C++
SQL



Libraries & Frameworks

TensorFlow

Keras

PyTorch

Scikit-learn

Pandas

NumPy

OpenCV

Tools

Git/GitHub

VS Code

Jupyter

Google Colab

AREAS OF EXPERTISE

Machine Learning

Deep Learning

Computer Vision

Neural Networks

Transfer Learning

Model Optimization

Image Processing

Data Analysis

EDUCATION

B.Sc. in AI Engineering 2023 – 2027

OSTİM Technical University

Ankara

GPA: 3.62

Focus: Neural Networks, Data Structures

LANGUAGES

Turkish (Native)

English (B2)



CERTIFICATIONS

Deep Learning Specialization

DeepLearning.AI (Andrew Ng) · 2026

REFERENCES

Ramin Abbaszadi

Lecturer at OSTİM Technical University

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Yasemin Arslan

Founder of Kairu